1. **Calcula y simplifica las siguientes expresiones:**

a. $\left(\frac{3}{2}-\frac{1}{5}+\frac{1}{10}\right)\cdot 5-\frac{3}{4}\cdot\frac{6}{5}=$

$=\left(\frac{15}{10}-\frac{2}{10}+\frac{1}{10}\right)\cdot 5-\frac{9}{10}=$

$=\left(\frac{14}{10}\right)\cdot 5-\frac{9}{10}=$

$=\frac{70}{10}-\frac{9}{10}=\frac{61}{10}$

b. $\left[\left(\frac{3}{2}-\frac{1}{5}\right)\cdot 5-\frac{1}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\left[\left(\frac{15}{10}-\frac{2}{10}\right)\cdot 5-\frac{1}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\left[\left(\frac{13}{10}\right)\cdot 5-\frac{1}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\left[\frac{13}{2}-\frac{1}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\left[\frac{65}{10}-\frac{1}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\left[\frac{64}{10}\right]\cdot\frac{3}{4}-\frac{6}{5}=$

$=\frac{24}{5}-\frac{6}{5}=\frac{18}{5}$

2. **Resuelve las siguientes expresiones:**

a.

$$\frac{1+\frac{1+\frac{1}{2}}{2}}{1-\frac{1-\frac{1}{3}}{3}}=$$

$$\frac{1+\frac{\frac{2}{2}+\frac{1}{2}}{2}}{1-\frac{\frac{3}{3}-\frac{1}{3}}{3}}=\frac{1+\frac{\frac{3}{2}}{2}}{1-\frac{\frac{2}{3}}{3}}=\frac{1+\frac{3}{4}}{1-\frac{2}{9}}=\frac{7/4}{7/9}=9/4$$

b.

$$\frac{\left(2+\frac{1}{3}\right)\cdot\left(4-\frac{2}{3}\right)}{1+\frac{5}{4}:\frac{3}{12}}=$$

$$\frac{\left(\frac{6}{3}+\frac{1}{3}\right)\cdot\left(\frac{12}{3}-\frac{2}{3}\right)}{1+\frac{5\cdot 12}{4\cdot 3}}=\frac{\left(\frac{7}{3}\right)\cdot\left(\frac{10}{3}\right)}{1+5}=\frac{70}{54}=35/27$$

3. Completa o resuelve en cada caso:

a. $\left[\left(\frac{3}{4}\right)^{[__]}\right]^4=\left(\frac{3}{4}\right)^{32}$

b. $\left(\frac{1}{4}\right)^2\cdot\left(\frac{1}{2}\right)^3=$

c. $\left(\frac{2}{7}\right)^{[__]}:\left(\frac{2}{7}\right)^3=\left(\frac{2}{7}\right)^6$

d. $\left(\frac{4}{5}\right)^6\cdot\left(\frac{4}{5}\right)^{[__]}=\left(\frac{4}{5}\right)^{14}$

a. $\left(\frac{3}{4}\right)^{32/4}=\left(\frac{3}{4}\right)^8$

b. $\left[\left(\frac{1}{2}\right)^2\right]^2\cdot\left(\frac{1}{2}\right)^3=\left(\frac{1}{2}\right)^{4+3}=\left(\frac{1}{2}\right)^7$

c. $\left(\frac{2}{7}\right)^{6+3}=\left(\frac{2}{7}\right)^9$

d. $\left(\frac{4}{5}\right)^{14-6}=\left(\frac{4}{5}\right)^8$

4. Calcula el resultado de las siguientes expresiones como una única potencia:

a. $\frac{3^2\cdot 2^5\cdot 3^{-1}}{9^4\cdot 6^2}=$

$=\frac{3^2\cdot 2^5\cdot 3^{-1}}{(3^2)^4\cdot(3\cdot 2)^2}=$

$=\frac{3^2\cdot 2^5\cdot 3^{-1}}{3^8\cdot 3^2\cdot 2^2}=$

$=\frac{3^2\cdot 2^5\cdot 3^{-1}}{3^8\cdot 3^2\cdot 2^2}=$

$=\frac{3^{2-1}\cdot 2^5}{3^{8+2}\cdot 2^2}=$

$=\frac{3^{2-1}\cdot 2^5}{3^{8+2}\cdot 2^2}=$

$=\frac{3\cdot 2^5}{3^{10}\cdot 2^2}=$

$= 2^3 \cdot 3^{-9} = \frac{2^3}{3^9}$

b. $\frac{\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^{-3} \cdot \left(\frac{81}{16}\right)^{-2}}{\left(\frac{3}{2}\right)^{-5} \cdot \frac{2}{3} \cdot \left[\left(\frac{2}{3}\right)^5\right]^2 \cdot \left(\frac{8}{27}\right)^3} =$

$\frac{\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^{-3} \cdot \left[\left(\frac{3}{2}\right)^4\right]^{-2}}{\left(\frac{3}{2}\right)^{-5} \cdot \frac{2}{3} \cdot \left[\left(\frac{2}{3}\right)^5\right]^2 \cdot \left[\left(\frac{2}{3}\right)^3\right]^3} = \frac{\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^{-3} \cdot \left(\frac{2}{3}\right)^8}{\left(\frac{2}{3}\right)^5 \cdot \frac{2}{3} \cdot \left(\frac{2}{3}\right)^{10} \cdot \left(\frac{2}{3}\right)^9} =$

$\frac{\left(\frac{2}{3}\right)^{5+0-3+8}}{\left(\frac{2}{3}\right)^{5+1+10+9}} = \frac{\left(\frac{2}{3}\right)^{10}}{\left(\frac{2}{3}\right)^{25}} = \left(\frac{2}{3}\right)^{-15} = \left(\frac{3}{2}\right)^{15}$

5. Opera y simplifica:

a. $\sqrt{2}+\sqrt{8}+\sqrt{18} - \sqrt{32} =$

$= \sqrt{2}+\sqrt{2^3}+\sqrt{3^2 \cdot 2} - \sqrt{2^5} =$

$= \sqrt{2}+2\sqrt{2}+3\sqrt{2} - 2^2\sqrt{2} =$

$=(6-4)\sqrt{2} =$

$=2\sqrt{2}$

b. $5\sqrt{2}+4\sqrt{8}+3\sqrt{18}+2\sqrt{32}+\sqrt{50} =$

$=5\sqrt{2}+4\sqrt{2^3}+3\sqrt{2 \cdot 3^2}+2\sqrt{2^5}+\sqrt{2 \cdot 5^2} =$

$=5\sqrt{2}+4 \cdot 2\sqrt{2}+3 \cdot 3\sqrt{2}+2 \cdot 2 \cdot 2\sqrt{2}+5\sqrt{2} =$

$=5\sqrt{2}+8\sqrt{2}+9\sqrt{2}+8\sqrt{2}+5\sqrt{2} =$

$=35\sqrt{2}$